NOT MEASUREMENT SENSITIVE

MIL-DTL-14639E
14 June 1999
SUPERSEDING
MIL-I-14639D
20 June 1987
MIL-I-3949D
31 December 1971

# DETAIL SPECIFICATION SHEET

# INSIGNIA, GRADE, ENLISTED PERSONNEL AND RANK, OFFICERS, METAL, GENERAL SPECIFICATION FOR

Inactive for new design after 25 January 1999.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-14639 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

## 1. SCOPE

1.1 Scope. This specification covers metal rank and grade insignia used by personnel of the Armed Forces (see 6.1).

1.2 Classification. The insignia covered by this specification will be as specified on the applicable military specification sheet (See 2.1 and 6.2).

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in preparing this document should be addressed to: Director, The Institute of Heraldry, US Army, 9325 Gunston Road, Room S112, Fort Belvoir, VA 22060-5579, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document.

AMSC N/A FSC 8455

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-F-495 - Finish, Chemical, Black for Copper Base Alloy

FEDERAL STANDARDS

FED-STD-595 - Colors Used in Government Procurement

SPECIFICATION SHEETS

See Supplement 1 for list of specification sheets)

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 42, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

STANDARD CHIP SETS

THE INSTITUTE OF HERALDRY, U.S. ARMY STANDARD HARD ENAMEL COLOR CHIP SET
THE INSTITUTE OF HERALDRY, U.S. ARMY STANDARD METAL FINISH CHIP SET

(The standard chip(s) for color or finish may be obtained from the procuring activity for Government procurements.)

DRAWINGS

*THE INSTITUTE OF HERALDRY*
B-3 (Series) - Insignia, Rank & Grade
B-13-12 - Attaching Devices, Heraldic, Prong and Clutch Type

(Figure 1 is a miniature copy of an Institute of Heraldry drawing and is for information only. Copies of specifications, standards, drawings, and specification sheets required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are issues of the documents cited in the solicitation (see 6.2)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-B-86 - Zinc-Alloy Die Casting
ASTM-B-487 - Metal and Oxide Coating Thickness by Microscopical Examination of a Cross Section, Measurement of
ASTM-D-2240 - Rubber Property - Durometer Hardness

AMERICAN SOCIETY FOR QUALITY CONTROL

ANSI/ASQC Z1.4m 1993 - Sampling Procedures and Tables for Inspection by Attributes

(Application for copies of ANSI/ASQC Z1.4 should be addressed to the American Society for Quality Control, 611 East Wisconsin Avenue, Milwaukee, WI 53202.)

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheets, the latter shall govern. The designs of the insignia shown on the specification sheets are for illustrative purposes only.

3.2 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmental preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.



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3.4 Materials. Materials shall as specified herein

3.4.1 Copper base alloy. The copper base alloy shall be roll polished, free from pits, scale (including red oxide), dents, nicks, cracks, scratches, segregation, and foreign inclusions that can not be removed in later processing. When tested as specified in 4.5.1, the chemical composition of the copper content of the copper base alloy shall be as specified in Table I.

TABLE I. **Copper content of copper base alloys.**

| ALLOY | COPPER |
|---|---|
| Yellow Brass | 64.0%-68.5% |
| Red Brass | 84.0%-86.0% |
| Gilding Metal | 94.0%-96.0% |
| Free Cutting Brass | 60.0%-63.0% |
| Bronze | 89.0%-91.0% |
| Nickel Silver | 63.0%-65.5% 1/ |
| Low Brass | 78.5%-81.5% |

1/ When nickel silver is used as a base material, the nickel content shall be not less than 18%.

3.4.2 Gold plating.

3.4.2.1 24 karat gold for plating. Gold for plating shall be 24 karat.

3.4.2.2. 22 karat gold for plating. When specified on the applicable specification sheet, gold for manufacturing insignia with nickel underplating shall be 22 karat.

3.4.3 Zinc aluminum alloy. The alloy used for die casting shall conform to alloy AG 40a or AC 41a of ASTM-B-86.

3.4.4. Nickel for plating. Nickel for plating shall be the nickel plating normally used in commercial practice.

3.4.5 Rhodium Plating. Rhodium for plating shall be the rhodium plating normally used in commercial practice.

3.4.6 Hard solder. Hard solder shall be a silver alloy solder having a melting point of not less than 1075 degrees Fahrenheit (561.7 degrees Celsius).

3.4.7 Enamel. All enamel shall be lead free.

3.4.7.1 Hard enamel. Hard enamel shall be a glass, vitreous type enamel fused with metallic oxides to produce the required color, opacity or translucence and shade.



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3.4.7.2 Epoxy resin (stoning epoxy). Epoxy resin shall be a stoning epoxy, have a shore D hardness of 85, incorporate an ultraviolet inhibitor, and be pigmented to produce the required color, opacity or translucence, and shade.

3.4.7.3 Baking enamel. The enamel for subdued insignia shall be a flat, synthetic baking enamel pigmented to match the standard established in FED-STD-595 as follows:

| COLOR | STANDARD CHIP NUMBER |
|---|---|
| Black | 37038 |
| Brown | 30099 |
| Olive Drab | 34096 |

3.4.7.4 Soft enamel. Soft enamel shall be a baking enamel pigmented to produce the required color or shade.

3.4.8 Lacquer. Lacquer shall be a pale clear synthetic lacquer. The use of a pigmented lacquer shall not be permitted.

3.5 Design. Except for Army enlisted grade insignia, the embossed design of all insignia shall be an exact duplicate of the design on the Government loaned hubs (see 3.8), from which the contractor's working dies shall be extracted. The contractor's working die shall be tooled and polished to remove any dents, nicks, scratches, or other imperfections.

3.5.1 Army enlisted grade insignia. Except for the Sergeant Major of the Army and the eagle for Specialist Four grade insignia, all Army enlisted grade insignia shall conform to the details and dimensions shown on TIOH drawing B-3-83. The Sergeant Major of the Army grade insignia and the eagle for the mirror-like polished Specialist Four grade insignia shall be an exact replica of a Government furnished hub (see 3.8) from which the contractor's working dies shall be extracted. The base of the Specialist Four grade insignia shall conform to the details and dimensions shown on TIOH drawing B-3-83A. When the subdued Specialist Four grade insignia is manufactured as a one piece item, the insignia shall conform to the details and dimensions shown on TIOH drawing B-3-83A. A forcer may be used to bring up the design of the eagle. The impression of the forcer shall be regular and symmetrical with the design of the insignia front. The insignia back shall have no sharp or rough edges or corners.

3.6 Construction. All insignia shall be constructed in accordance with the applicable specification sheet and the requirements specified herein. Either stamping or die casting methods of manufacturing shall be acceptable for subdued insignia. Prongs may be cast as an integral part of the insignia when die cast method is used. Unless otherwise specified on the applicable specification sheet, a tolerance of ± 0.005 inch shall be allowed on all interior dimensions. All mirror-like polished insignia shall be stamped.

3.6.1 Stamping, trimming, and piercing. The stamping shall have a well defined die struck edge. The insignia shall be trimmed and, when applicable, pierced to the die struck edge. All edges shall be clean, smooth and free from burrs, drag, step, and rough edges.


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The stamping, piercing, and trimming operations shall not damage or distort the design or alter the shape of the item.

3.6.2 Die casting. When die cast method is applicable, the item shall be free from porosity, blow holes, or other casting defects. Before finishing, all flash lines shall be removed and the edges of the item shall be smooth. Items that are die cast from zinc aluminum specified in 3.4.3 shall be copper plated. When tested as specified in 4.5.2.2, the plating shall be a minimum of 0.0003 inch thick.

3.6.3 Soldering and electronic fusion. Unless otherwise specified on the applicable specification sheet, all soldering shall be accomplished using hard solder specified in 3.4.6 or by electronic fusion. Joints shall be clean, smooth, strong, and free from burned or reduced areas. There shall be no excess solder, and all excess flux shall be removed. When hard solder is used, the soldered parts and prongs shall not separate at the joint when tested as specified in 4.5.4. When fused joints are used, the prong shall not separate at the joint when tested as specified in 4.5.5.

3.6.4 Forcers. When forcers are permitted to bring up the design, the back of the insignia shall have no sharp or rough edges. The impression of the forcer shall be symmetrical with the embossed design of the insignia. Sufficient metal shall be left near any trimmed area to assure clean, smooth trimming without distortion. Unless otherwise specified on the applicable specification sheet, mock forcers shall not be permitted. When mock forcers are permitted, the forcer shall be used only on the areas specified.

3.6.5 Attaching devices. Attaching devices shall be as specified on the applicable specification sheet.

3.6.5.1 Prong (registering). Prongs shall be made from nickel silver specified in 3.4.1. The prongs shall be of the dimensions shown on the applicable specification sheet. They shall be positioned and attached as required by the applicable specification sheet. Unless otherwise specified, the prong shall be secured to the insignia by soldering or electronic fusion.

3.6.5.2 Integral prong. When prongs are die cast as integral parts of the insignia and tested in accordance with 4.5.2.1, no evidence of breaking or cracking shall occur.

3.6.5.3 Prong and wing-type clutch. Prongs shall be made from nickel silver specified in 3.4.1. Unless otherwise specified on the applicable specification sheet, the prongs as shown on TIOH drawing B-13-12 (Figure 1) shall be 5/16 inch ± 1/64 inch long. The prongs shall be hard soldered or electronically fused in the location specified in the applicable specification sheet. Wing-type clutches as shown on TIOH drawing B-13-12 (Figure 1) shall be made from any type brass. When tested as specified in 4.5.8.2 it shall not be possible to remove the clutch from the prong without first depressing the release mechanism.

3.6.5.4 Prong with flat ball-type clutch. Prongs shall be made from nickel silver specified in 3.4.1. Unless otherwise specified on the applicable specification sheet, the prongs as shown on TIOH drawing B-13-12 (Figure 1) shall be 5/16 inch ± 1/64 inch long. The prongs shall be hard soldered or electronically fused in the location specified in the



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applicable specification sheet. Flat ball-type clutches as shown on TIOH drawing B-13-12 (Figure 1) shall be made from any type brass. When tested as specified in 4.5.8.3 it shall not be possible to remove the clutch by hand without pulling up on the ball.

3.6.5.5 Rivets. Rivets used for attaching a superimposed device to the front of the base shall be made from any type brass, shall be solid or tubular, and shall be swaged to insure a tight and close fit between the component parts. At the option of the manufacturer, rivets may be soldered to or be an integral part of the device.

3.7 Finish. Unless otherwise specified, the front and edges (surfaces visible from the front) shall be finished as specified on the applicable specification sheet and shall match the TIOH standard enamel color or metal finish chips as required (see 2.2.2). All brass, mirror-like polished insignia shall be lacquered. When plating is specified, the entire insignia, excluding attaching devices, shall be plated. The backs of brass, gold, nickel, and rhodium plated insignia shall be clean and bright. Except for Navy and Marine Corps insignia, gold plated insignia may be lacquered at the option of the manufacturer. All Navy and Marine Corps gold plated insignia shall be lacquered. When an insignia is lacquered, the front surface and edges shall be lacquered using lacquer specified in 3.4.8. Lacquering of the entire insignia is optional.

3.7.1 Subdued insignia. The face and edges of all subdued insignia shall be coated with enamel specified in 3.4.7.3. Prior to enameling the entire insignia shall be given a black chemical finish conforming to MIL-F-495. When the subdued insignia is die cast from zinc aluminum specified in 3.4.3, the insignia shall be copper plated prior to finishing. When tested as specified in 4.5.1, the plating shall be a minimum of 0.003 inch thick. When tested as specified in 4.5.7.2, the enamel shall not flake or chip.

3.7.2 Plating. All plating shall be by electroplating methods. The plating shall be non-porous and continuous over the entire plated surface. There shall be no cut-through, shaded, peeled, or blistered plating. Plating shall be smooth, fine grained, adherent, and free from pits, nodules, porosity, indications of burning, excessive edge build up, and other detrimental defects. Unless otherwise specified, plating of attaching devices is not required.

3.7.2.1 24 karat gold plating. Gold plating shall be accomplished using gold specified in 3.4.2.1. Prior to plating, all surfaces shall be cleaned of all matter that may affect the gold plating. The use of nickel as an undercoating with 24 karat gold shall not be permitted. When tested as specified in 4.5.3, no visible chemical reaction such as evolution of gases shall appear.

3.7.2.2. 22 karat gold plating. The gold plating shall be accomplished using gold specified in 3.4.2.2. Prior to plating, all surfaces shall be cleaned of all matter that may affect the gold plating. The use of nickel as an undercoating shall be permitted. When tested as specified in 4.5.3, no visible chemical reaction such as evolution of gases shall appear.

3.7.2.3 Nickel plating. Nickel for plating shall be the nickel specified in 3.4.4 and shall be not less than 0.0003 inch thick. Testing shall be standard commercial. In case of a dispute, testing shall be in accordance with ASTM-B-487.



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3.7.2.4 Nickel underplating. When nickel underplating is specified, nickel specified in 3.4.4 shall be used and shall be not less than 0.00025 inch thick. Testing shall be standard commercial. In case of a dispute, testing shall be in accordance with ASTM-B-487.

3.7.2.5 Rhodium Plating. When rhodium plating is specified, rhodium specified in 3.4.5 shall be used. When tested as specified in 4.5.3, no chemical reaction shall occur. When rhodium plating is required over brass, nickel underplating specified in 3.4.4 shall be required.

3.7.3 Enameling. Unless otherwise specified on the applicable specification sheet, all enameling shall be accomplished using hard enamel specified in 3.4.7.1 or epoxy specified in 3.4.7.2. The enamel or epoxy shall be applied so as to be uniform in color, free from bubbles, foreign inclusions, cracking, crazing, or other defects which might affect appearance. There shall be no overrunning of enamel or epoxy.

3.7.3.1 Colors. Enamel or epoxy colors shall match the shades of the Institute of Heraldry hard enamel color chips specified on the applicable specification sheet (see 2.2.2).

3.7.3.2 Hard enameling. Hard enamel specified in 3.4.7.1 shall be charged, fired, and stoned level with the dikes. No design elements shall be removed during the stoning process. There shall be no noticeable burn spots as a result of the firing. The hard enamel shall then be polished to produce a glass like finish. The finish shall be adherent and free from bubbles, pits, foreign inclusions, cracking, crazing, burned edges, or darkened enamel.

3.7.3.3 Epoxy resin. Epoxy resin specified in 3.4.7.2 shall be applied so as to be uniform in color, free from bubbles, foreign inclusion, or other defects which affect appearance. The epoxy shall be stoned level with the dikes and polished to produce a glass like finish. No design elements shall be removed during the stoning process. When tested as specified in 4.5.7.1, the epoxy shall have a shore D hardness of not less than 85.

3.7.3.4 Baking enamel. The face and edges of subdued insignia shall be coated with enamel specified in 3.4.7.3. Prior to enameling, the en;tire insignia shall be given a black chemical finish conforming to MIL-F-495. When tested as specified in 4.5.7.2, the baking enamel shall not flake or chip.

3.7.3.5. Soft enamel. Soft enamel shall be as specified in 3.4.7.4. When tested as specified in 4.5.7.3, no color shall be transferred to the cotton, and the enameled surface shall remain unaffected except for a slight loss of luster.

3.7.4 Cutting down. When necessary to remove nicks, scratches, pin holes, or other blemishes, the item shall be cut down by means of grease or oil buffing. The insignia shall not be cut down to the extent that any details of the design are obliterated or reduced.

3.7.5 Lacquering. The front, back, and edges of all brass items shall be coated with clear lacquer specified in 3.4.8. Gold plated and rhodium plated items may be coated with clear lacquer at the option of the manufacturer. The lacquer film shall be continuous, level,


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adherent, and free from lint, dust, and other foreign matter. When tested as specified in 4.5.6 the tissue paper shall not adhere to the clear lacquered surface.

3.8 Government loaned property. Hubs will be loaned by the Government and shall be used to make the contractor's working dies necessary for one contract or order (see 6.4).

3.9 Marking for identification. The contractor shall stamp his trademark or other identifying mark and the letters "GI" (GI for government procurement only) legibly and inconspicuously on the back of each item. When practical, the items will be stamped in the forced areas. In addition, precious metal insignia shall bear a quality marking.

3.10 Marking for cleaning. Marking for cleaning of gold plated items shall be as follows: "To clean, rub gently with a soft, dry cloth. Do not clean with a chemically treated cloth or abrasive."

3.11 Workmanship. The finished item shall be clean, well made, and shall conform to the acceptable quality levels established by this specification.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, all inspection shall be performed in accordance with the test conditions specified in 4.4 and 4.5.

4.3 First article inspection. Inspection and testing of the first article (see 3.2) shall be made of a completely finished item for all provisions of this specification applicable to the end product examination and tests.

4.4 Conformance inspection. Inspection shall be in accordance with the provisions set forth herein, except where otherwise indicated (see 6.3). Sampling for inspection shall be in accordance with ANSI/ASQC Z1.4.

4.4.1 Testing of components. Inspection shall be performed on components and materials listed in Table II for the test characteristics shown therein. Material listed in Table II may be accepted on a contractor's certificate of compliance for requirements specified in applicable paragraphs of this specification. The government reserves the right to test samples of gold (without additional charge) to verify the contractor's certification.



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TABLE II. **Testing of components.**

| COMPONENT | CHARACTERISTIC | RQM'T ¶ | TEST METHOD | RESULTS REPORTED AS |
|---|---|---|---|---|
| Copper base alloy | material identification | 3.4.1 | 4.5.1 | nearest 0.1% |
| Zinc aluminum alloy | material identification | 3.4.3 | standard commercial | pass/fail |
| Rhodium plating | material identification | 3.4.5 | standard commercial | pass/fail |
| Enamel (hard) | material identification | 3.4.7.1 | standard commercial | pass/fail |
| Enamel (soft) | material identification | 3.4.7.4 | standard commercial | pass/fail |
| Enamel (baking) | material identification | 3.4.7.3 & 3.7.3.4 | 4.5.7.2 | pass/fail |
| Epoxy resin | material identification | 3.4.7.2 | standard commercial | pass/fail |
| Gold for plating | material identification & karat | 3.4.2.1 & 3.4.2.2 | standard commercial | pass/fail |
| Lacquer | material identification | 3.4.8 | standard commercial | pass/fail |
| Nickel for plating | material identification | 3.4.4 & 3.7.2.3 | standard commercial | pass/fail |
| Nickel for underplating | material identification | 3.4.4 & 3.7.2.4 | standard commercial | pass/fail |
| Nickel silver | material identification | 3.4.1, 3.6.5.1 3.6.5.3, & 3.6.5.4 | standard commercial | nearest 0.1% |
| *Solder, hard* | *material identification* | 3.6.3 | standard commercial | pass/fail |

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operation or assemblies listed in 3.5, 3.6, and 3.7 are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance

4.4.2 End item inspection.

4.4.2.1 Visual examination of insignia. Visual examination shall be made using Table III. Examination shall be made at a distance of approximately 16 to 22 inches with illumination equal to average daylight and arranged so as to avoid as much reflected light as possible. Defects designated by and asterisk (*) shall be classified as major when seriously affecting appearance or serviceability and minor when affecting appearance or serviceability but not seriously. The product unit for the examinations shall be one completely fabricated insignia. Any dimension which is not within the specified tolerance shall be classified as a defect.



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TABLE III. **Defects.**

| EXAMINE | DEFECT | CLASSIFICATION | | |
|---|---|---|---|---|
| | | MAJOR | (*) | MINOR |
| Metal color and finish | Poor match to the TIOH Standard Metal Finish Chip | X | | |
| | Not required finish | | * | |
| | Discoloration, such as spot, stain, or speck | | * | |
| | Not clean | | | * |
| | Paired insignia not matching in color, shade and finish (when applicable | | * | |
| | Background or recessed areas not finished as specified | X | | |
| | Surface contains pits, scale, nick, scratch, crack, machine mark, pin hole, segregation, rupture, foreign matter, or other blemish | | * | |
| | Buff drag, blushed, or cloudy finish | | * | |
| Hard enamel and epoxy | Not enameled (when required) | X | | |
| | Incorrect color or shade | X | | |
| | Cracked, chipped, or crazed | | * | |
| | Not stoned smooth, i.e., is coarse, uneven, or has drag marks | | * | |
| | Not level with dikes, i.e., contains high or low spots | | * | |
| | Bubbles, blisters, holes, lumps, or foreign inclusions | | * | |
| | Laps, overruns, or skips | | * | |
| | Blemished, i.e., spot, stain, spatter, burned area, lack of gloss | | * | |
| Soft/baking enamel | Not enameled or area of no enamel | X | | |
| | Incorrect color or shade | X | | |
| | Area of thin enamel | | | X |
| | Baking enamel not flat and dull | X | | |
| | Not adherent, i.e., blistered, flaking, or peeling | | * | |
| | Foreign matter imbedded | | * | |
| | Discoloration | | * | |
| | Coating not continuous and level, i.e., runs, drips, or drops | | * | |
| Design | Construction or design details obscured, altered, or not conforming to Government hub | X | | |
| | Not dapped (when required) | X | | |
| | Any significant detail not clear, altered, reduced, distorted, or missing | X | | |


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TABLE III. **Defects** – Continued.

| EXAMINE | DEFECT | CLASSIFICATION | | |
|---|---|---|---|---|
| | | MAJOR | (*) | MINOR |
| | Any warp, twist or distortion, irregular surface contour or outline ..... | | * | |
| | Any detail struck over resulting in a double impression ..... | X | | |
| | Any area not pierced (when required) ..... | X | | |
| Plating | Not plated (when required) ..... | X | | |
| | Not type specified ..... | X | | |
| | Plating is not smooth, continuous, or adherent ..... | X | | |
| | Foreign matter imbedded ..... | | * | |
| Workmanship and construction | Surface or edge not clean, not smooth or free from burrs, roughness, drag, step, or tool marks ..... | | * | |
| | Metal marks on exposed surface (such as nick, dent, dig, gouge, or scratch) ..... | | * | |
| | Die-struck edge, rim, or outline is not defined when applicable ..... | X | | |
| | Not trimmed to the die-struck edge, when applicable ..... | | * | |
| | Pierced edge not clean and smooth, i.e., any burr, drag, cutter, or file mark ..... | | * | |
| | Design or shape is damaged, distorted, or altered by piercing operation ..... | X | | |
| | Forcer used when not specified or insignia forced in other than specified location ..... | | * | |
| | Metal torn or broken out by use of forcer .... | | * | |
| | Any component missing, bent, twisted, broken, or malformed ..... | | * | |
| | Solder does not completely unite parts, i.e., any perceptible opening between component parts ..... | | * | |
| | Any area burned or reduced in soldering ... | | * | |
| | Soldered joint not clean or smooth; excess solder or excess flux not removed ..... | | | X |
| | Back not flat or smooth when required ..... | | | X |
| | Superimposed design not securely riveted to insignia ..... | | * | |
| Attaching device | Not specified type, size, or material ..... | X | | |
| | Not attached in specified manner ..... | X | | |
| | Any component missing ..... | X | | |
| | Defective, i.e., any part damaged or malformed affecting use ..... | | * | |



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TABLE III. **Defects** – Continued.

| EXAMINE | DEFECT | CLASSIFICATION | | |
|---|---|---|---|---|
| | | MAJOR | (*) | MINOR |
| | Defective, does not operate as intended .... | | * | |
| | Not positioned as specified ..... | X | | |
| | Clutch does not engage prong or is loose fit | X | | |
| | Clutch pierced through by prong ..... | | * | |
| | Prong loose ..... | X | | |
| Identification | Missing, incorrect, illegible, misspelled, not accomplished as specified, or not placed as required ..... | | * | |
| Lacquer | Not lacquered when required ..... | X | | |
| | Areas of no lacquer ..... | | * | |
| | Foreign matter embedded in finish, i.e., lint, dust, etc. ..... | | * | |
| | Hazy, rainbow effect, cloudy, or powdering | | * | |
| | Not smooth, continuous or adherent, i.e., flaking, blistering, peeling, rippling, or has run, sag, or drops ..... | | * | |
| | Not set to touch, i.e., tacky when pressure is applied to coating ..... | X | | |
| Quality of unplated metal | Surface pitted, porous, crystalline, spotted, opened grained, or otherwise impaired ... | X | | |

4.4.2.2 End item testing. Testing of the completely fabricated item shall be performed in accordance with Table IV for the characteristics shown therein. The sample unit shall be one completely fabricated badge, badge clasp, bar, or pendant. The sample unit for the test for hard soldered or fused joints shall be one finished item. All items shall be tested and one defect shall be scored regardless of how many prongs on that item failed. Individual tests are detailed in 4.5.


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TABLE IV. **Testing of end item.**

| CHARACTERISTIC | RQM'T ¶ | TEST METHOD | RESULTS REPORTED AS |
|---|---|---|---|
| Test for copper content | 3.4.1 | 4.5.1 | pass/fail |
| Tests for plating | | | |
| Nickel plating | 3.7.2.3 | standard commercial | pass/fail |
| Nickel underplating not permitted with 24 karat gold plating | 3.7.2.1 | standard commercial | pass/fail |
| Nickel underplating with 22 karat gold plating | 3.7.2.2 | standard commercial | pass/fail |
| Copper plating | 3.7.1 | standard commercial | pass/fail |
| Acid test: | | | |
| 24 karat gold plating | 3.7.2.1 | 4.5.3 | pass/fail |
| 22 karat gold plating | 3.7.2.2 | 4.5.3 | pass/fail |
| Rhodium plating | 3.7.2.5 | 4.5.3 | pass/fail |
| Test for lacquer: | 3.7.5 | 4.5.6 | pass/fail |
| Tests for joints: | | | |
| Hard soldered joints | 3.6.3 | 4.5.4 | pass/fail |
| Electronically fused joints | 3.6.3 | 4.5.5 | pass/fail |
| Tests for attaching devices: | | | |
| Integral prong | 3.6.5.2 | 4.5.2.1 | pass/fail |
| Prong with wing-type clutch | 3.6.5.3 | 4.5.8.2 | pass/fail |
| Prong with flat ball clutch | 3.6.5.4 | 4.5.8.3 | pass/fail |
| Tests for enamels: | | | |
| Epoxy resin | 3.7.3.3 | 4.5.7.1 | pass/fail |
| Baking enamel | 3.7.3.4 | 4.5.7.2 | pass/fail |
| Soft enamel | 3.7.3.5 | 4.5.7.3 | pass/fail |

4.4.2.3 Examination of Captain (Navy) or Colonel insignia for pairing. Any Captain (Navy) or Colonel insignia mismatched shall be classified as a defect. The sample unit for this examination shall be one pair of Captain (Navy) or Colonel insignia.



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4.4.2.4 Examination of paired insignia for compatible finish. Any paired insignia having finish and/or color which is not closely matched to each other shall be classified as a defect. The sample unit for this examination shall be one pair of insignia mounted on a card.

4.5 End item tests.

4.5.1 Copper content test for copper base alloy. Copper content shall be determined by standard commercial testing. Results shall be evaluated to determine compliance with the requirements specified in 3.4.1.

4.5.2 Test of Die Casting.

4.5.2.1 Test for integral prongs. Insignia to be tested shall be placed in a jig which allows no movement of the insignia and positions prongs at a 30 degree angle above the horizontal. A vertical 10 pound load shall be applied to the top third of each prong. The prongs shall be examined to determine compliance with 3.6.5.2.

4.5.2.2 Direct micro thickness test for copper plating. Copper plating thickness shall be determined by standard commercial testing. Results shall be evaluated to determine compliance with 3.6.2.

4.5.3 Acid test for gold and rhodium plating. If the test insignia has been lacquered, the lacquer shall be completely removed prior to testing with an organic solvent. The test acid shall be applied as follows: Place a drop of acid of not less than 1/16 inch in diameter on three (3) different spots on the plated surface allowing the drops to remain for not less than one (1) minute during which time the surface of the item shall be inspected to determine compliance with the requirements in 3.4.2.1 and 3.4.2.2. No visible chemical reactions shall occur. Acid drops to be placed on flat surfaces on the obverse face when possible. A minimum of two spots must withstand the acid test. The test acid shall be applied at room temperature (60 to 80° Fahrenheit/15.6 to 26.7° Celsius) and shall consist of a solution containing 50% by volume of chemically pure nitric acid (specific gravity 1.42) and an equal volume of distilled water.

4.5.4 Test for hard soldered joints. Hard soldered items shall be placed in an oven maintained at 1075° Fahrenheit ± 5° Fahrenheit (561.7° Celsius ± 2.78° Celsius) for 15 minutes. While at this temperature the item shall be lifted by the attaching device and shall be inspected to determine compliance with the requirements specified in 3.6.3.

4.5.5 Test for electronically fused joints. Items to be tested shall be anchored on a horizontal surface. Each prong shall be grasped at least 1/3 its length above the fused joint and bent through an angle of 90 degrees (45 degrees to each side of the vertical) until the prong breaks. The fused joint shall then be examined to determine compliance with 3.6.3. A bending tool in the form of a 45 degree template may be used for this test provided the prong is gripped at least 1/3 its length above the fused joint.

4.5.6 Test for lacquer. At room temperature (60 to 80° Fahrenheit/15.6 to 26.7° Celsius), press a piece of tissue paper against the lacquered surface for 15 seconds, using



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any pressure capable of being exerted between thumb and two fingers, after which the pressure shall be released and the item inspected to determine compliance with 3.10.

4.5.7 Test for enamels.

4.5.7.1 Test for shore hardness of epoxy resin. Shore hardness shall be determined in accordance with ASTM-D-2240. Results shall be evaluated to determine compliance with the requirements of 3.7.3.3.

4.5.7.2 Test for baking enamel. Draw a knife edge across the face side of the specimen making four parallel scores 1/16 inch apart. Repeat this process drawing the knife edge this time perpendicular to the previously scored lines, making small blocks in the finish. The finish shall then be examined to determine compliance with 3.7.1.

4.5.7.3 Acetone test for soft enamel. The enameled surface shall be wiped five times with a piece of cotton saturated with acetone. The cotton and enamel shall then be examined to determine compliance with 3.4.7.4

4.5.8 Test for attaching device.

4.5.8.1 Test for integral prongs. Insignia to be tested will be placed in a jig which allows no movement of the insignia and positions prongs on a 30 degree angle above the horizontal. A vertical 10 pound load shall be applied to the top third position of each prong. The prongs will be examined to determine compliance with 3.6.5.2.

4.5.8.2 Test for prong and wing-type clutch. The clutches shall be removed and replaced ten times from the prong using the clutch release mechanism. An attempt shall then be made to remove the clutch by hand without first depressing the release mechanism. An inspection shall be made at this time of the clutch and prong to determine compliance with the requirements specified in 3.6.5.3.

4.5.8.3 Test for prong and flat ball clutch. The clutches shall be removed and replaced from the prong by manually pulling up on the ball of the clutch ten times. An attempt shall then be made to remove the clutch by hand without first grasping the ball. An inspection shall be made at this time of the prong with flat ball clutch to determine compliance with the requirements specified in 3.6.5.4.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.



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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Insignia covered by this specification are intended to be worn by officers of the Army, Navy, Marine Corps, and Air Force and by enlisted personnel of the Army and Air Force on designated uniforms to indicate the rank or grade of the wearer.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. The title, number, and date of the applicable specification sheet (see 1.2 and 2.1).

d. When a first article is not required (see 3.2).

e. Packaging requirements (see 5.1).

f. When special marking is required (see 3.10).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first production sample or a standard production item from the contractor's current inventory as specified in 4.4. The first article should consist of one completed item. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.8.

6.5 Subject term (key word) listing.

Acetone
Enamel, baking
Enamel, hard
Epoxy
Gold
Lacquering
Lead



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Nickel
Nickel silver
Nitric acid
Plating
Prong, integral
Prong, registering
Rhodium
Soldering
Zinc aluminum

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - IH
Air Force - 45
Navy - NU

Review activities:
Air Force - 32, 82, 99
Navy - MC
Coast Guard - CG
DLA - CT

Preparing activity:
Army - IH

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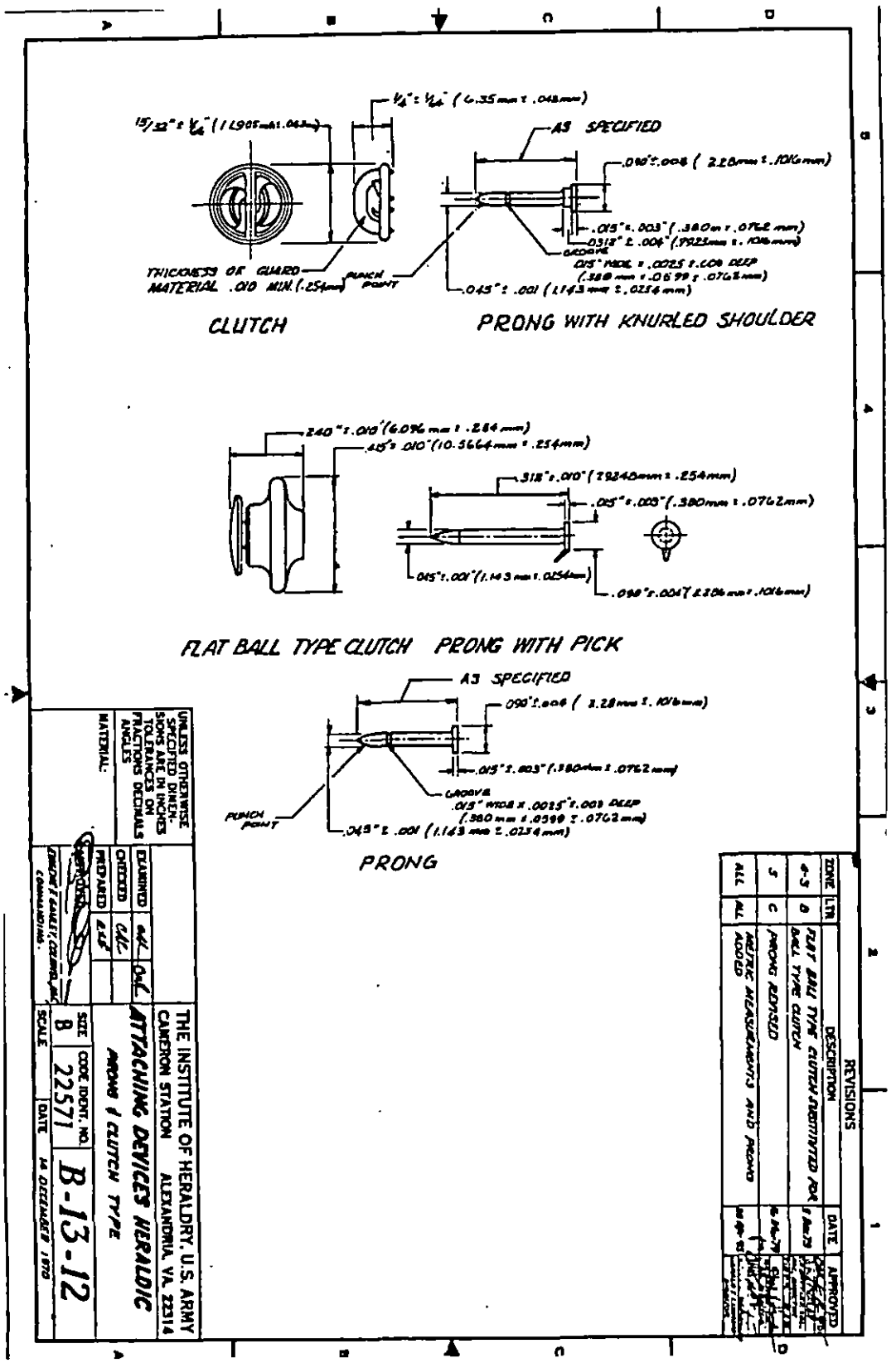



FIGURE 1. **Attaching devices heraldic, pin & clutch type.**



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-DTL-14639E | 2. DOCUMENT DATE *(YYYYMMDD)*<br>19990614 |
|---|---|---|

3. DOCUMENT TITLE
INSIGNIA, GRADE, ENLISTED PERSONNEL AND RANK, OFFICERS, METAL, GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME *(Last, First, Middle Initial)* | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS *(Include ZIP Code)* | d. TELEPHONE *(Include Area Code)*<br>(1) Commercial<br>(2) DSN<br>*(If applicable)* | 7. DATE SUBMITTED *(YYYYMMDD)* |

8. PREPARING ACTIVITY

| a. NAME<br>The Institute of Heraldry, U.S. Army | b. TELEPHONE *(Include Area Code)*<br>(1) Commercial: (703) 806-4982<br>(2) DSN: 656-4982 |
|---|---|
| c. ADDRESS *(Include ZIP Code)*<br>Technical and Producton Division<br>9325 Gunston Raod, Bldg 1466, Rm. S-112<br>Fort Belvoir, VA 22060-5579 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Standardization Program Office (DLSC-LM)<br>8725 John J. Kingman Road, Suite 2533<br>Fort Belvoir, Virginia 22060-6221<br>Telephone (703) 767-6888 DSN 427-6888 |

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